create_pin_guide

NAME

create_pin_guide
Creates a pin guide in the current design.

SYNTAX

```
int create_pin_guide
       -boundary { { {llx lly} {urx ury} } |
                   { {x y} {x y} {x y} {x y} ... } } |
                   geometric_objects
       [-layers layer_list]
       [-parents cell_list]
       [-name pin_guide_name]
       [-exclusive]
       [-pin_spacing number_of_wire_tracks]
       [-cell cell]
       object_list
```

Data Types

llx	float
lly	float
urx	float
ury	float
x	float
y	float
geometric_objects	collection
layer_list	list
cell_list	list
pin_guide_name	string
number_of_wire_tracks	int
cell	string or collection
object_list	list

ARGUMENTS

-boundary { {{llx lly} {urx ury}} | {{x y} {x y} {x y} {x y}...} } | geometric_objects

Specifies the boundary coordinates of the pin guide. The boundary can be a rectangle or a polygon. A rectangle is specified by its lower-left and upper-right coordinates, for example, {llx lly} {urx ury}. A polygon is specified by its points, for example, {x y} {x y} {x y} {x y} and so on.

Polygons can also be specified as the combined area of a mixed collection of objects with physical geometry, such as poly_rects, geo_masks, shapes, layers, and other physical objects. In the case of poly_rects, geo_masks, shapes, or other physical objects, the resulting area will include the areas of each object. In the case of layers, the resulting area will include the area of every shape in the layer.

When specifying the boundary polygon as a collection of physical objects, the resulting area must resolve to a single, connected polygon. It is an error to specify a collection of objects with a combined area that resolves to multiple polygons.

-layers layer_list

Specifies a list of layers for the pin guide. Pins will be placed only on these layers. If layers are not specified, pins can be placed on any layer.

-parents cell_list

Specifies a list of block cells to which this pin guide applies. If not specified, this pin guide is applied to the current design.

-name pin_guide_name

Specifies the name of the pin guide.

-exclusive

Specifies that this pin guide is exclusive. This prevents the placement of terminals within the pin guide, except for pins, pins of nets, and ports associated to this pin guide.

-pin_spacing number_of_wire_tracks

Specifies the number of wire tracks between pins within this pin guide. When this option is specified, the pin placement command will place pins, within the pin guide, such that the pin minimum spacing is equivalent to the number of specified wire tracks.

-cell cell
Specifies the physical cell where the pin guide is to be added. The pin guide is created in the cell's reference block using the coordinate system of the cell argument's top block. The cell must reference a block, not a library cell, unless this command is executed in the library manager. In the library manager, pin guides can be created in library cells. When not specified, the pin guide is created in the current block.

object_list
Specifies a list of physical pins nets, or bundles of nets to associate with this pin guide if the parents for the pin guide are block cells. Specifies a list of physical ports, nets, or bundles of nets to associate with this pin guide if the parent for the pin guide is the current design. The object list must be homogeneous; it cannot contain a mixture of different object types. During pin assignment, the terminals generated for these objects are constrained to the pin guide bounding box. Currently bundles of supernets is not supported. If bundle of supernets is specified then it will be skipped with a warning message.

DESCRIPTION

The create_pin_guide command enables you to create a pin guide with the specified boundary in the current design. The pin guide has one or more parent cells, to which the pin guide is applied. The parents of the pin guide can be the current design or a list of block cells. During pin assignment, the terminals generated for these objects are constrained to the pin guide bounding box.

If the parent of the pin guide is the current design, the pin guide can be associated with a list of physical ports or nets. If the parents of the pin guides are block cells, the pin guide can be associated with a list of physical pins or nets. The list must be homogeneous; it cannot contain a mixture of object types. Each pin, net, and port can be associated to multiple pin guides.

EXAMPLES

The following example creates a pin guide named 'PB_1' with rectangular boundary that applies to all layers of the current design. It is associated with physical ports of the current design.

```
prompt>  create_pin_guide -boundary {{100 100} {200 200}} \
   -name PB_1 [get_ports clk*]
```

The following example creates a pin guide with a rectilinear boundary that applies to the M1 and M2 layers only in the current design. It is associated with physical nets in the current design.

```
prompt>  create_pin_guide -boundary {{0 0} {2000 0} {2000 2000} \
   {1000 2000} {1000 1000} {0 1000}} -layers {M1 M2} [get_nets n*]
```

The following example creates a pin guide with a rectangular boundary that is associated with physical pins of the macro cell 'macroA'.

```
prompt>  set macro macroA
prompt>  create_pin_guide -boundary {{100 100} {200 200}} \
   -parents [get_cells $macro] [get_pins $macro/*]
```

SEE ALSO

add_to_pin_guide(2)
get_pin_guides(2)
remove_from_pin_guide(2)
remove_pin_guides(2)
report_pin_guides(2)

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